NEF ABUNDANCE: FRAMEWORK – JET SEPARATION IN GENERAL DIMENSION

1. Jet separation

Fix an integer $d \geq 2$. We use the following source inputs, denoted collectively by (I_d).

- For every projective klt $\mathbb{Q}$-pair (Y, B) of dimension $j < d$ and every nef rational Cartier divisor M, if $K_Y + B$ is pseudo-effective and, for some $a \in \mathbb{Q}_{>0}$,

$$(K_Y + B + aM) \cdot C > 0$$

  for every integral curve through a sufficiently very general point, then $K_Y + B + cM$ is big for every $c \in \mathbb{Q}_{>0}$.
- Every projective log canonical $\mathbb{Q}$-pair of dimension at most d with nef adjoint has semiample adjoint.
- Every projective $\mathbb{Q}$-factorial generalized klt $\mathbb{Q}$-pair of dimension at most d, with fixed nef rational Cartier b-divisor data and pseudo-effective adjoint, admits a terminating choice of minimal-model program with a nef endpoint. The program extracts no divisors, preserves $\mathbb{Q}$-factorial generalized klt models, and on a common resolution the initial adjoint equals the final adjoint plus an effective divisor exceptional over the final model.

The first input is the lower-dimensional assertion (P_j) of [1, Section 3.1]. The last two are the companion inputs used in [1, Section 2]; here they are assumptions, not independently established higher-dimensional theorems.

Let X be a smooth projective variety over $\mathbb{C}$ of dimension d, with $K_X \sim 0$ and $q(X) = 0$. Let D be a nef Cartier divisor such that $\mathrm{n}(X, D) = d$ and $D^d = 0$. Assume (I_d).

Setup.

- Fix an ample Cartier divisor A.
- Fix $\epsilon \in \mathbb{Q}_{>0}$ sufficiently small. For integers $n \geq 1$, put

$$a_n = ((D + A/n)^d)^{1/d}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(D + A/n).$$

  Then P_n is ample and rational Cartier, $r_n \to \infty$, and $P_n^d \to \epsilon^d$. Indeed, $a_n \to 0$ and $|r_n a_n - \epsilon| \leq a_n/2$ for all sufficiently large n. The symbol n indexes the sequence; the dimension is d.

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- Let $p_i : X \times X \to X$ be the projections. Put

$$Z = \mathbb{P}_{X\times X}\big(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D)\big), \qquad \pi : Z \to X \times X.$$

  Let S_1, S_2 be the two summand sections and set $U = Z \setminus (S_1 \cup S_2)$. Thus $\dim Z = 2d + 1$.

- Set $\xi = c_1(\mathcal{O}_Z(1))$. For an integer $s > 0$, put

$$R_{n,s} = (p_1 \circ \pi)^* P_n + (p_2 \circ \pi)^* P_n + s\xi.$$

  We use $\pi_*\mathcal{O}_Z(a) = \mathrm{Sym}^a(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D))$ for $a \geq 0$. Whenever $|kR_{n,s}|$ is used, $k > 0$ is an integer with kP_n Cartier.

- Choose ϵ sufficiently small that

$$(14\epsilon)^d < \tfrac{1}{4}, \qquad 40\epsilon < \tfrac{3}{8}.$$

  The integer s is chosen after ϵ and before n tends to infinity. Any jet-separating multiple is chosen after n.

Jet separation. Let W be smooth, let L be a Cartier divisor, and let $z \in W$. For $j \geq 0$, $|L|$ separates j-jets at z if

$$\forall u \in \mathcal{O}_{W,z}(L) \quad \exists \sigma \in H^0(W, \mathcal{O}_W(L)) \quad \sigma_z - u \in \mathfrak{m}_z^{j+1}\mathcal{O}_{W,z}(L).$$

Lemma 1.1. *There exists a positive integer s such that, for every $n \gg 0$, $|kR_{n,s}|$ separates $(2d+2)k$-jets at a very general point $z \in U$, for all sufficiently divisible positive integers k.*

Draft status. This is a conditional, source-based proof framework following [1, Sections 4–6]. The source inputs (I_d) and the inherited moving-centre and Frobenius arguments have not been independently verified here. No unconditional higher-dimensional nonvanishing or abundance theorem is asserted.

2. Auxiliary estimates

The geometric argument will produce subvarieties with bounded degree and bounded canonical intersection. We first explain how these bounds contradict a growing lower bound for the degrees of curves. We then record the moving-center results from [9] that supply the required subvarieties. All varieties in this section are over $\mathbb{C}$.

2.1. **The bounded-degree obstruction.**

Lemma 2.1. *For every integer $f \geq 1$ there is an integer $m_f > 0$ with the following property. Let Y be a smooth integral projective variety of dimension f, let D_0 be a reduced simple normal crossing divisor or the zero divisor, and let L_0 be a nef integral Cartier divisor of full nef dimension on Y. Suppose that P_0 is a nef rational Cartier divisor and that r, C_0, C_1 are positive real numbers satisfying*

$$0 < P_0^f \leq C_0, \qquad P_0 - rL_0 \text{ is nef.}$$

If $A_0 = K_Y + D_0 + L_0$ is big and

$$A_0 \cdot P_0^{f-1} \leq C_1 P_0^f, \tag{1}$$

then

$$r \leq (m_f C_1)^{f-1} C_0. \tag{2}$$

The integer m_f depends only on f, and not on the variety or the divisors. For dimensions $1 \leq f \leq d$, one may replace all m_f by a single integer m depending only on d.

Proof. The pair (Y, D_0) is log canonical, its boundary coefficients belong to $\{0, 1\}$, and its nef polarizing divisor L_0 has Cartier index one. By [2, Theorem 1.3], an integer m_f depending only on these data makes $|m_f A_0|$ birational. Here A_0 is integral Cartier, so no rounding or additional denominator is involved.

Resolve the base ideal by a projective birational morphism $\mu : \widetilde{Y} \to Y$, with $\widetilde{Y}$ smooth, and write

$$\mu^*(m_f A_0) = H + F,$$

where H is the basepoint-free moving part and F is effective. The morphism defined by H is birational onto its image; in particular H is nef and big. In the rest of this proof, P_0 and L_0 denote their pullbacks to $\widetilde{Y}$. Effectivity of F and nefness of P_0 give

$$H \cdot P_0^{f-1} \leq m_f C_1 P_0^f. \tag{3}$$

For $f \geq 2$, put $s_i = P_0^{f-i} \cdot H^i$ for $0 \leq i \leq f$. Both divisors are nef and big, so the s_i are positive. The Khovanskii–Teissier inequalities [8, Corollary 1.6.3(i) and Example 1.6.4] $s_i^2 \geq s_{i-1} s_{i+1}$ show that the ratios s_i / s_{i-1} decrease. Hence

$$P_0 \cdot H^{f-1} \leq P_0^f \left(\frac{H \cdot P_0^{f-1}}{P_0^f} \right)^{f-1} \leq (m_f C_1)^{f-1} C_0. \tag{4}$$

Choose a very general point $y \in \widetilde{Y}$ over the isomorphism opens of both μ and the morphism defined by H. We also choose $\mu(y)$ in the very general locus that tests full nef dimension of L_0. Pull back $f - 1$ general hyperplanes through the image of y under

the morphism defined by H. This subsystem has only the base point y, so the general members meet properly and define an effective curve cycle of class H^{f-1}; the component C through y is a curve because their local equations are independent at y. Nefness gives

$$P_0 \cdot C \leq P_0 \cdot H^{f-1}.$$

The curve C is not contracted by μ, and its image passes through the chosen very general point. Full nef dimension and integrality therefore give $L_0 \cdot C \geq 1$. Since $P_0 - rL_0$ is nef, we obtain $P_0 \cdot C \geq r$, which proves (2).

If $f = 1$, take $C = Y$ directly. Then $P_0 \cdot Y \geq rL_0 \cdot Y \geq r$, and the required bound is $r \leq C_0$. Finally, [2, Theorem 1.3] applies to every multiple of its prescribed integer. A common multiple of the finitely many integers $m_1, \ldots, m_d$ therefore gives the last assertion. □

In particular, the hypotheses of Lemma 2.1 cannot hold along a sequence with $r \to \infty$ and with C_0, C_1 bounded, even if the varieties and divisors vary. The following observation explains how full nef dimension and pseudo-effectivity of the canonical divisor pass to the moving subvarieties used below.

Lemma 2.2. *Let X be a normal integral projective variety that is not uniruled, and let L be a nef Cartier divisor of full nef dimension on X. There is a complement X° of a countable union of proper closed subsets of X with the following property. If Y is a smooth integral projective variety and $g : Y \to X$ is generically finite onto a positive-dimensional image meeting X°, then g^*L has full nef dimension, Y is not uniruled, and K_Y is pseudo-effective.*

Proof. Choose X° so that every curve through any of its points has positive L-degree and no rational curve meets X°. The first condition is the very general characterization of full nef dimension. For the second, rational curves are parametrized by countably many algebraic families; the closure of the evaluation image of each such family is proper because X is not uniruled.

Since $g(Y)$ meets X°, it is contained in none of the closed sets excluded in its definition. A very general point of Y thus maps into X° and lies in the quasi-finite locus of g. Every curve through such a point has a curve as its image, so its g^*L-degree is positive. This proves full nef dimension. A covering family of rational curves on Y would similarly give a nonconstant rational curve meeting X°, a contradiction. The pseudo-effectivity of K_Y now follows from [4]. □

2.2. **Subadjunction and high-multiplicity components.** We use the following five results from the section *Tools for moving base components* of [9]. Their hypotheses distinguish generic information along a center from global information about the ambient

variety. Intersections on an integral subvariety are computed after pullback to its normalization and a smooth projective resolution. A base ideal is *isolated at the generic point* of a subvariety when its localization there is primary for the maximal ideal.

Lemma 2.3. *Let Z be a projective $\mathbb{Q}$-factorial klt variety, let $\Theta \geq 0$ be a rational divisor, and let $V \subset Z$ be integral of dimension $f > 0$. Suppose that (Z, Θ) is log canonical at the generic point of V and has a divisor of log discrepancy zero with center V. If P is a nef rational Cartier divisor on V and $\rho : V^* \to V$ is a smooth projective resolution factoring through its normalization, then*

$$K_{V^*} \cdot (\rho^* P)^{f-1} \leq (K_Z + \Theta)|_V \cdot P^{f-1}.$$

Only generic log canonicity is required. The proof in [9] uses the dlt modification of [6, Theorem 2.10], whose truncated boundary leaves an effective surplus supported over the non-lc locus. On a minimal stratum over V, adjunction gives a pair that is klt over the generic point of its connected-fiber base. This suffices for the klt-trivial canonical bundle formula and its b-nef moduli divisor [5, Definition 3.1 and Theorem 3.5]. The effective discriminant, the moduli divisor, and finite ramification have nonnegative intersections with the pulled-back nef class. The comparison uses canonical Weil cycles, so it requires no $\mathbb{Q}$-Gorenstein assumption on the normalization of V.

Lemma 2.4. *Let Z be a projective $\mathbb{Q}$-factorial klt variety of dimension e, let R be a nef rational Cartier divisor, and let $k > 0$ be an integer such that kR is Cartier. Let $0 \neq \mathcal{H} \subset H^0(Z, \mathcal{O}_Z(kR))$ be a linear subspace with proper base locus and base ideal $\mathfrak{b}$. Suppose that V is an integral base-locus component of dimension $f < e$, isolated at its generic point η_V, and that $\eta_V \in Z_{\mathrm{sm}}$. Let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}_{Z,\eta_V}$. If $\mathfrak{b}\mathcal{O}_{Z,\eta_V} \subset \mathfrak{m}^h$ for an integer $h > 0$, there are a rational number $0 < c \leq (e-f)k/h$ and an effective rational divisor $\Theta \sim_{\mathbb{Q}} cR$ such that (Z, Θ) is log canonical at η_V and has a log canonical place centered on V. The degree and canonical bounds are*

$$\begin{aligned} R^f \cdot V &\leq (k/h)^{e-f} R^e, \\ K_{V^*} \cdot R^{f-1} &\leq (K_Z + cR)|_V \cdot R^{f-1} \qquad (f > 0), \end{aligned} \tag{5}$$

where V^* is a smooth projective resolution. The degree bound also holds for $f = 0$.

The degree estimate is the local multiplicity bound h^{e-f} combined with successive general cuts by the given linear series. The boundary is obtained by averaging general members with total coefficient equal to the generic log canonical threshold of $\mathfrak{b}$. Thus it is an actual divisor to which Lemma 2.3 applies.

2.3. **Full jet kernels and restriction to a fiber.**

Lemma 2.5. *Let Z be an integral projective variety of dimension e, let R be a nef, big, semiample rational Cartier divisor, and fix an integer $k > 0$ such that kR is Cartier. Choose positive real numbers $\tau_0 < \cdots < \tau_e$ with consecutive gap $\delta > 0$. For $x \in Z_{\mathrm{sm}}$, put*

$$\mathcal{H}_i(x) = H^0\big(Z, \mathcal{O}_Z(kR) \otimes \mathfrak{m}_x^{\lceil k\tau_i \rceil}\big).$$

Suppose that for very general x all these spaces are nonzero and the base locus of $\mathcal{H}_0(x)$ has a positive-dimensional component through x. If $k\delta \geq 4$, then, after an algebraic parameter extension and restriction to nonempty opens, there are an integral parameter variety T with a dominant marked-point map $x : T \to Z_{\mathrm{sm}}$, a fixed index $i \in \{0, \ldots, e-1\}$, and a family of integral subvarieties V_t containing $x(t)$ and common to the base loci of $\mathcal{H}_i(x(t))$ and $\mathcal{H}_{i+1}(x(t))$. In particular, the incidence

$$\Gamma = \{(t, z) : z \in V_t\} \subset T \times Z$$

dominates Z.

For general t, the dimension f of V_t lies in $[1, e-1]$, V_t is an isolated component of the higher base locus at its generic point, and the entire higher-kernel base ideal is contained there in the ordinary power $\mathcal{I}_{V_t}^h$, where

$$h = \lceil k\delta/2 \rceil, \qquad k/h \leq 2/\delta.$$

The incidence base ideal is likewise contained in $\mathcal{I}_\Gamma^h$ on a dense open of Γ. These containments are taken in the smooth ambient open. If Z is also $\mathbb{Q}$-factorial and klt, then

$$\begin{aligned} R^f \cdot V_t &\leq (2/\delta)^{e-f} R^e, \\ K_{V_t^*} \cdot R^{f-1} &\leq (K_Z + cR)|_{V_t} \cdot R^{f-1}, \qquad 0 < c \leq 2(e-f)/\delta. \end{aligned} \tag{6}$$

The top intersection on V_t is positive when V_t meets the open on which the big semiample contraction is an isomorphism.

The use of the full kernels matters: differentiation in the parameter lowers vanishing at the moving mark and still gives a global section in the lower kernel. Incidence dominance makes these parameter directions span the normal directions. The vanishing of all derivatives of order less than h then gives the ordinary-power containment, which can be restricted to general fibers as follows.

Lemma 2.6. *In the setting of Lemma 2.5, let $g : Z \to B$ be a morphism. There is a dense open of the incidence with the following property. Choose (t, z) in this open, put $b = g(z)$, and suppose that V_t is smooth over a smooth open of its actual image at z. Let F be the reduced component through z of the scheme fiber of $V_t \to g(V_t)$. Near z in*

Z_b, the restricted higher-kernel base ideal has support exactly F and is contained in the ordinary power $\mathcal{I}_F^h$.

If F is proper in an integral ambient fiber component containing it, the restricted series is nonzero there and F is an isolated base component at its generic point. Applying Lemma 2.4 on that ambient component additionally requires its projectivity, $\mathbb{Q}$-factoriality, klt singularities, and smoothness at the generic point of F.

Here the fiber is chosen through a general incidence point. The statement does not assert the same conclusion for an arbitrary special fiber. Smoothness over the actual image identifies the restricted ideal of V_t with the reduced ideal of F locally; this is what preserves ordinary powers.

The $\mathbb{Q}$-factoriality needed for the base-component estimates can also be supplied by a small projective $\mathbb{Q}$-factorialization $\sigma : Z' \to Z$ of a klt ambient variety. This passage is made only for components that meet the isomorphism open of σ. Their strict transforms retain the generic base-ideal and multiplicity conditions for the pulled-back series, and the testing class $\sigma^* R$ is nef. Since $K_{Z'} = \sigma^* K_Z$, projection formula gives the same degree and canonical-intersection bounds on a common resolution. The pullback testing class need not be ample; nefness is the property used in these estimates. Passing to higher resolutions does not change the canonical intersections, because their additional exceptional divisors have zero intersection with the pulled-back testing class to the required power.

2.4. **Finite covers with a fixed branch complement.** A bound on the degree of a branched cover does not make the possible covers finite. The final companion input supplies the needed fixed branch complement from a family whose branch divisors do not sweep the base.

Lemma 2.7. *Let Y be a normal integral projective variety, let $\mathcal{V} \to T$ be a smooth projective family with integral fibers over an integral parameter variety, and let $g : \mathcal{V} \to Y$ be a morphism. Suppose that the general fiber maps $g_t : V_t \to Y$ are dominant and generically finite of degree at most a fixed integer e.*

After a finite parameter extension and shrinking, spread the geometric generic components of the zero divisor of the relative Jacobian over Y_{sm} to divisors $\mathcal{R}_1, \dots, \mathcal{R}_N$ in $\mathcal{V}$. Retain this notation after the base change. Let I consist of the indices j for which $\overline{g_t((\mathcal{R}_j)_t)}$ is a divisor in Y for general t. Assume that

$$\overline{g(\mathcal{R}_j)} \neq Y \qquad (j \in I).$$

After shrinking again, there is a smooth nonempty open $Y^\circ \subset Y$ over which every general finite normal Stein cover of g_t is finite étale. Only finitely many such normal covers of Y

occur, up to isomorphism over Y. The conclusion holds simultaneously for finitely many projections; the open and the finite lists may depend on all fixed parameters of the family.

To see the role of the hypothesis, remove Y_{sing} and the finitely many proper image closures $\overline{g(\mathcal{R}_j)}$ for $j \in I$. Every branch prime of a general Stein cover is accounted for by one of these relative Jacobian components. Purity therefore makes the restricted covers étale. The fundamental group of the resulting smooth complex variety is finitely generated, so it has only finitely many covers of degree at most e. Finally, a finite normal cover of Y is determined by its function field, hence by its restriction to this fixed open.

Lemma 2.8. *Assume* (I_d). *Let (Y, B) be a projective klt $\mathbb{Q}$-pair of dimension at most d, with $T = K_Y + B$ pseudo-effective and $\kappa(Y, T) > 0$. Let M be nef and rational Cartier. If $(T + aM) \cdot C > 0$ for every integral curve through a sufficiently very general point, for some $a > 0$, then $T + cM$ is big for every rational $c > 0$.*

Proof. Apply the source reduction for positive Iitaka dimension [1, Proposition 3.3]. Its model-program and ordinary abundance inputs are included in (I_d); its fibre dimensions are strictly smaller than d. This reduction is retained as a source-dependent input here. □

3. Geometric estimates

In the technical arguments below, put $L = D$, $t = 1/n$, $r = r_n$, $P = P_n$, and $q = s$. The letter L keeps the source notation for the fixed nef Cartier divisor; q is the bundle parameter.

$$P = r(L + tA), \qquad r \to \infty, \qquad P^d \to \epsilon^d. \tag{7}$$

For each fixed q, discard finitely many terms so that

$$\epsilon^d/2 \le P^d \le 2\epsilon^d, \qquad r > q + 1. \tag{8}$$

Every curve through a very general point of X has positive integral L-degree, hence P-degree at least r. Thus a covering family of curves of bounded P-degree will give a contradiction.

We specify how the same obstruction applies to subvarieties and covers. A resolution $W \to X$ has an effective canonical divisor: pull back a nowhere-zero canonical form on the smooth locus of X and use terminality. Hence W, and therefore X, is not uniruled [4]. By Lemma 2.2, if a smooth projective variety Y maps generically finitely to a positive-dimensional subvariety containing a sufficiently very general point of X, then Y is not uniruled and the pulled-back Cartier divisor L_Y has full nef dimension. In particular,

K_Y is pseudo-effective. If $f = \dim Y < d$, the induction hypothesis (P_f) applied to $(Y, 0)$ gives

$$K_Y + L_Y \text{ big.} \tag{9}$$

All occurrences of "very general" below include the fixed conditions of Lemma 2.2.

Consequently, for $f < d$ it suffices to find on such a Y a nef testing divisor P_Y with

$$0 < P_Y^f \le C_0, \qquad P_Y - rL_Y \text{ nef}, \qquad K_Y P_Y^{f-1} \le C P_Y^f,$$

where C_0, C are independent of t. Indeed $L_Y P_Y^{f-1} \le P_Y^f/r$, so Lemma 2.1 applies. We will also use that lemma in dimension d, after proving the required bigness separately. For the rest of this section, a bound is called uniform if it is independent of t; it may depend on parameters fixed before t tends to zero. Since the dimensions lie in a finite range, Lemma 2.1 excludes all such configurations once r is sufficiently large.

3.1. **The projective bundle and its volume.** We now seek a jet system whose order is much larger than the scalar bound. The additional direction comes from the fibres of a projective bundle; its two summands keep track of the two copies of X. For a divisor on X, subscripts $1, 2$ denote pullback from the two factors of $X \times X$. Set

$$Z = \mathbb{P}\big(\mathcal{O}_X(L)_1 \oplus \mathcal{O}_X(L)_2\big) \xrightarrow{\pi} X \times X, \qquad \xi = c_1(\mathcal{O}_Z(1)), \qquad R = P_1 + P_2 + q\xi, \tag{10}$$

where q is a positive integer to be fixed shortly. We use the convention $\pi_*\mathcal{O}_Z(a) = \mathrm{Sym}^a(\mathcal{O}_X(L)_1 \oplus \mathcal{O}_X(L)_2)$ for $a \ge 0$. The tautological class ξ is nef, and R is ample. The product and projective-bundle descriptions show that Z is klt. The two natural sections, called the axes, have classes $\xi - L_1$ and $\xi - L_2$. The canonical bundle formula gives

$$K_Z \sim -(\text{sum of the two axes}). \tag{11}$$

We call the complement of the axes the torus open.

The fibre of either projection $Z \to X$ over a smooth point is, up to a constant one-dimensional factor,

$$Z_{\mathrm{sl}} = \mathbb{P}(\mathcal{O}_X \oplus \mathcal{O}_X(L)), \qquad R_{\mathrm{sl}} = P + q\xi.$$

Its axes have classes ξ and $\xi - L$, and its canonical divisor is again minus their sum. The slice is projective and klt. In particular it has the singularity hypotheses needed to apply the base-component estimates, using a small $\mathbb{Q}$-factorialization if necessary as explained after Lemma 2.5.

Write $e = 2d + 1$. There are positive constants depending only on d such that, for each fixed q and all sufficiently small t,

$$c_n q\epsilon^{2d} \le R^e \le C_n q\epsilon^{2d}, \qquad R_{\mathrm{sl}}^{d+1} \le C_{\mathrm{sl},d} q\epsilon^d. \tag{12}$$

To see this, expand in nef classes and use

$$\pi_*(\xi^j) = \sum_{a=0}^{j-1} L_1^a L_2^{j-1-a} \quad (j \geq 1).$$

The term containing exactly one ξ is $(2d+1)\binom{2d}{d} q (P^d)^2$, giving the lower bound. For the upper bound, every occurrence of L_i can be bounded in mixed nef intersections by P_i/r. After factoring out q, the remaining powers of q/r are bounded by one once $r \geq q$. There are only finitely many terms depending on d. The slice calculation is the same, using $\pi_*(\xi^j) = L^{j-1}$. Together with (8), this proves (12).

For a nef rational Cartier divisor R and a smooth point z, its *Seshadri constant* is

$$\varepsilon(R; z) = \inf_{C \ni z} \frac{R \cdot C}{\operatorname{mult}_z C},$$

where C ranges over integral curves through z. Thus a lower bound for this constant controls degree relative to multiplicity at the point.

Proposition 3.1. *For a suitable fixed positive integer q, and all sufficiently small t in (7), there is a smooth point $z \in Z$ in the torus open, lying over smooth points of both factors, such that*

$$\varepsilon(R; z) > 2d + 2. \tag{13}$$

More generally, for any prescribed dense open subsets U_1, U_2 of the smooth locus of X, the point may be chosen over $U_1 \times U_2$. In particular, these opens may be the isomorphism loci of fixed resolutions of X.

We prove the proposition in the remainder of this section. Choose a positive rational gap δ such that

$$(2/\delta)^n C_{\mathrm{sl},d} \epsilon^d < 1. \tag{14}$$

Next choose q so large that $e+1$ increasing levels of gap δ, with lowest level strictly greater than $2d+2$, lie strictly below $(c_n q \epsilon^{2d})^{1/e}$. These choices precede the limit $t \to 0$.

Fix the prescribed opens U_1, U_2, if any. Suppose that (13) fails throughout their smooth torus preimage for arbitrarily small t. For any such t, asymptotic Riemann–Roch and the elementary jet count at a smooth point show that all the full kernels at the chosen levels are nonzero for sufficiently large divisible k. Indeed the highest level has e-th power smaller than R^e. Failure of (13) supplies, through each very general point, a curve with degree divided by multiplicity smaller than the lowest level. Every section in the lowest kernel vanishes identically on that curve: otherwise its intersection with the curve is at least its vanishing order times the curve's multiplicity, contradicting the degree inequality.

Thus the lowest base locus has a positive-dimensional component through the mark. Take $k\delta \geq 4$ and apply Lemma 2.5.

We obtain a family of proper integral components $V \subset Z$ whose incidence dominates Z. For its general members, of dimension f, the estimates are

$$\begin{aligned} &0 < R^f \cdot V \leq (2/\delta)^{e-f} R^e, \\ &K_{V^*} R^{f-1} \leq (K_Z + cR)|_V R^{f-1}, \qquad 0 < c \leq 2(e-f)/\delta. \end{aligned} \tag{15}$$

Here and below classes on a resolution are understood by pullback. The positivity follows from ampleness of R. General members meet the torus open, so (11) makes the ambient canonical contribution nonpositive. All constants in these bounds are uniform now that q, ϵ, δ have been fixed.

3.2. **Excluding positive-dimensional slice fibres.** We first constrain the two projections of V to X. For either projection, consider the component F of a general fibre of $V \to \operatorname{im}(V \to X)$ through a general incidence point. Choose that incidence point in the opens of Lemma 2.6, over very general points of both factors of X, and in the smooth torus open. We claim that

$$\dim F \notin \{1, \ldots, d\}. \tag{16}$$

Suppose instead that $1 \leq f_{\mathrm{sl}} = \dim F \leq d$. The ambient fibre is Z_{sl}, of dimension $d+1$. The restriction lemma gives a nonzero restricted subseries with F an isolated base component and base ideal contained generically in $\mathcal{I}_F^h$, where $h = \lceil k\delta/2 \rceil$. Generic smoothness is imposed over the actual image of V; this is why a general fibre through the incidence point was chosen. The point is also smooth in the ambient slice. Applying Lemma 2.4 on that slice gives

$$\begin{aligned} &0 < R_{\mathrm{sl}}^{f_{\mathrm{sl}}} \cdot F \leq (2/\delta)^{d+1-f_{\mathrm{sl}}} R_{\mathrm{sl}}^{d+1}, \\ &K_{F^*} R_{\mathrm{sl}}^{f_{\mathrm{sl}}-1} \leq (K_{Z_{\mathrm{sl}}} + cR_{\mathrm{sl}})|_F R_{\mathrm{sl}}^{f_{\mathrm{sl}}-1}, \end{aligned} \tag{17}$$

with $c \leq 2(d+1-f_{\mathrm{sl}})/\delta$. Cartier multiples are chosen before restriction, so the constant line from the fixed factor causes no change in this calculation.

Let $W_0 \subset X$ be the image of F in the other factor. There are three cases: F is generically finite over a proper subvariety of X; F contains the projective-line fibres over its image; or F is a multisection over all of X.

A generically finite map to a proper subvariety. If $F \to W_0$ is generically finite and $\dim W_0 < d$, use $P_Y = R_{\mathrm{sl}}|_{F^*}$ and the pullback L_Y from X. The first inequality in (17) gives a uniform positive volume bound. Since F meets neither axis generically, its ambient canonical term is nonpositive. Hence the second inequality gives a uniform canonical-intersection ratio. Moreover $P_Y - rL_Y$ is nef. The image W_0 contains the very general

projection of the incidence point, so (9) applies. Lemma 2.1 excludes this case for large r.

A projective-line bundle over a proper subvariety. If $F \to W_0$ is not generically finite, its general fibre has dimension one. Since the ambient projection is a projective-line bundle and F is integral, F is the full bundle over W_0. Write $w = \dim W_0 = f_{\mathrm{sl}} - 1$. If $w = 0$, then F is a projective-line fibre and $R_{\mathrm{sl}} \cdot F = q$. The first inequality in (17), together with (12) and (14), gives

$$q \le (2/\delta)^n R_{\mathrm{sl}}^{d+1} \le (2/\delta)^d C_{\mathrm{sl},d} q \epsilon^d < q,$$

which is impossible.

Suppose $w > 0$. Resolve W_0 to W_0^* and take as a resolution of F the induced projective bundle $p_0 : F^* \to W_0^*$. Its relative canonical divisor is $-2\xi + p_0^* L$, exactly the restriction of $K_{Z_{\mathrm{sl}}}$. Cancelling this term in (17) gives

$$p_0^* K_{W_0^*} R_{\mathrm{sl}}^w \le c R_{\mathrm{sl}}^{w+1} \cdot F. \tag{18}$$

The divisor $K_{W_0^*}$ is pseudo-effective by the very general point condition. On expanding the left side and pushing down, its term with exactly one ξ is $wqK_{W_0^*}P^{w-1}$; every other nonzero term is nonnegative, since it pairs the pseudo-effective canonical divisor with nef classes. On the right, the same expansion used for (12) gives

$$(w+1)qP^w \cdot W_0 \le R_{\mathrm{sl}}^{w+1} \cdot F \le C'_n q P^w \cdot W_0, \tag{19}$$

using $r \ge q$. The slice degree bound and the first inequality in (19) bound $0 < P^w \cdot W_0$ uniformly. Equations (18)–(19) also give

$$K_{W_0^*} P^{w-1} \le \frac{cC'_d}{w} P^w \cdot W_0.$$

Apply (9) and Lemma 2.1 on W_0^* with the pullbacks of P, L. This excludes the second case.

A multisection over X. The only remaining possibility is that $f_{\mathrm{sl}} = d$ and $F \to X$ is generically finite of degree $e > 0$. Intersect F with each of the two Cartier axes and push the resulting cycles down to X. This gives effective integral Weil divisors D_1, D_2. The axes do not contain F, and their classes differ by L; projection of the associated rational section, or equivalently its norm, gives

$$D_1 - D_2 \sim \pm eL. \tag{20}$$

For either axis S, the classes $R_{\mathrm{sl}} - qS$ and $R_{\mathrm{sl}} - P$ are nef. Intersecting with the effective cycle $S \cdot F$ and then using nefness on F yields

$$qD_i P^{d-1} \le qS \cdot F \cdot R_{\mathrm{sl}}^{d-1} \le R_{\mathrm{sl}}^d \cdot F. \tag{21}$$

Thus both axis-image degrees are uniformly bounded.

Set $G = D_1 + D_2$. It is $\mathbb{Q}$-Cartier because X is $\mathbb{Q}$-factorial. Choose $b > 0$ rational sufficiently small that (X, bG) is klt. We claim

$$\kappa(K_X + bG) > 0. \tag{22}$$

The Iitaka dimension is at least zero, since $K_X + bG \sim_{\mathbb{Q}} bG$. If it were zero, the model-program and ordinary abundance inputs in (I_d) would give a finite ordinary MMP to a good model Y with transformed adjoint rationally linearly trivial. The transform of K_X is linearly trivial as well. Consequently the effective transform of G is zero: all its components have been contracted. On a common resolution $a : U \to X$, $b_Y : U \to Y$, relation (20) then expresses a^*L up to rational linear equivalence as a signed b_Y-exceptional divisor E. Divisors exceptional over X are also exceptional over Y, since the MMP extracts none. Nefness of a^*L and the negativity lemma imply $E \leq 0$. Its class is also pseudo-effective. A nonzero anti-effective divisor has negative intersection with a sufficiently high power of an ample divisor, so $E = 0$. This contradicts full nef dimension of L, and proves (22).

The already proved positive-Iitaka-dimension case (Proposition 2.8) now gives

$$K_X + bG + L \text{ big.}$$

Its full-positivity hypothesis holds because L has full nef dimension and $K_X + bG$ is pseudo-effective. Take a log resolution $a : Y \to X$ of G and let D_0 be the reduced simple normal crossings divisor consisting of the strict support of G and every exceptional divisor. Decrease b if necessary; Proposition 2.8 still applies. Terminality and the support of a^*G then give

$$K_Y + D_0 + a^*L \;\geq\; a^*(K_X + bG + L)$$

up to rational linear equivalence and an effective difference. Indeed the finitely many exceptional discrepancies are positive, and decreasing b makes each coefficient of $b\,a^*G$ no larger than the corresponding coefficient of $K_Y - a^*K_X + D_0$. Hence the left side is big.

Test on Y with a^*P. Exceptional divisors contribute zero to intersection with $(a^*P)^{d-1}$, and $K_X \sim 0$. The degree of the reduced strict support is no larger than GP^{d-1}, bounded by (21); also $LP^{d-1} \leq P^d/r$. Since (8) bounds P^d away from zero, we obtain a uniform constant C such that

$$(K_Y + D_0 + a^*L)(a^*P)^{d-1} \leq C(a^*P)^d.$$

Lemma 2.1, now with its reduced boundary D_0, gives the final contradiction. This proves (16).

Thus a general moving component V has fibre dimension either zero or $d+1$ over its image in either factor. Fibre dimension $d+1$ would make V the full inverse image of its image in that factor. Since V is proper, the image has dimension at most $d-1$, so the other projection would have fibre dimension between 1 and d, already excluded. Both projections of V are therefore generically finite onto their images, and $\dim V \leq d$. If $\dim V < d$, the bounds (15), the nef class $R - rL_i$, and (9) again contradict Lemma 2.1. Only d-dimensional correspondences, dominant over both factors, remain.

3.3. **Correspondences and their branch divisors.** For these remaining V, (15) gives uniform bounds

$$R^d \cdot V \leq C, \qquad K_{V^*} R^{d-1} \leq cR^d \cdot V \leq C, \tag{23}$$

after increasing C. Each projection has bounded degree: $R - P_i$ is nef, and P^d is bounded below.

For each fixed t, resolve the general members in their algebraic family and shrink the parameter space so that the resolutions form a smooth projective family with integral fibres and morphisms to X in both slots. The dominant incidence and all numerical bounds persist. Finite extensions of the parameter space are allowed. Over X_{sm}, name the components of the relative Jacobian divisors as in Lemma 2.7. Consider a component whose image on a general member is a divisor $J \subset X$.

Such divisors J cannot sweep X. If they did, choose a general one through a very general point. Let E_J be a ramification divisor over it on the corresponding V^*. Pullback of a canonical form from X gives an effective canonical divisor on V^*. Regularity follows by factoring through a resolution of X and using its canonical singularities; the coefficient of E_J is positive because its image meets X_{sm} generically. The difference $R - P_i$ is nef, so (23) and projection formula bound

$$0 < e_J := P^{d-1} \cdot J \leq C.$$

On a resolution J^*, Lemma 2.3, applied to (X, J) at the generic point of J, gives

$$\begin{aligned} K_{J^*} P^{d-2} &\leq (K_X + J) \cdot J \cdot P^{d-2} \\ &= J^2 P^{d-2} \leq \frac{(JP^{d-1})^2}{P^d} \leq C' e_J. \end{aligned} \tag{24}$$

The pair is lc at that generic point because X is normal and hence smooth there. The middle inequality is the Hodge index inequality, applied on a resolution and obtained by ample approximation of the pulled-back testing class. It does not require J to be nef. The final constant is uniform by the volume lower bound and the bound for e_J.

Since J sweeps X, its resolution satisfies the very general point conditions used in (9). With testing class $P|_{J^*}$, whose top self-intersection is e_J, Equation (24) gives precisely

the hypotheses of Lemma 2.1 in dimension $d-1$. This contradiction excludes sweeping branch divisors.

Lemma 2.7 now gives, for each slot and this fixed t, only finitely many possible finite normal Stein covers of X, up to isomorphism over X. The point of the preceding step is that degree bounds alone would not give this conclusion: non-sweeping branch images provide a fixed smooth branch complement for the family. Neither that complement nor the resulting finite lists need be uniform in t.

3.4. **Fixed covers and an abelian model.** Each V^* maps birationally to both of its normal Stein covers, and hence determines a graph of a birational map between a pair of covers in the finite lists. We explain why some such fixed pair has a family of graphs dominating its product. Restrict the finite set of pairs to those realized by at least one original member V^* satisfying (23). The original incidence dominates Z, so the projected incidences for these pairs cover a dense open of $X \times X$. For a fixed pair, graphs of birational maps lie in countably many Hilbert schemes. Stratify these schemes so that the fibres are integral and flat and both projections are birational; the latter conditions may be tested by dominance and degree one after shrinking. Because $\mathbb{C}$ is uncountable, a dense open of $X \times X$ cannot be covered by countably many proper closed subsets. Thus, for one fixed pair T_1, T_2, some integral graph family has dominant evaluation to $X \times X$. Since $T_1 \times T_2$ is finite over $X \times X$ and has the same dimension, this evaluation is dominant onto $T_1 \times T_2$ as well.

Both covers are non-uniruled because they are finite over X. The birational-group theorem of Hanamura [7, Theorems 2.1–2.2], in the form recalled in [3, Proposition 3.7 and Theorems 3.8–3.9], implies that they are birational to an abelian variety. Here are the details of the application. Identify the two covers birationally by one map in the family. On a suitable smooth projective birational model of the resulting non-uniruled variety, the reduced birational group is a group scheme locally of finite type, with identity component an abelian variety acting regularly. Conjugate the graph family to this model and shrink again to obtain a flat family of integral graphs. By the flat-graph representability statement, its reduced integral parameter space maps to the reduced birational group. Its connected image lies in one translate of the identity component. Product-dominant evaluation therefore gives a dense orbit for that abelian identity component. The orbit is a quotient of an abelian variety by a stabilizer, hence is itself an abelian variety. The fixed cover is birational to it. This is the birational-group part of the fixed-cover argument in [9, Proposition 6.11]; no canonical nonvanishing conclusion from that proposition is being used.

This structural conclusion depends only on the two covers. We may therefore return to an original member V^* belonging to this pair, which retains (23), and resolve further to obtain a birational morphism

$$p_0 : V^* \longrightarrow A_0$$

to an abelian variety. Higher resolutions do not change the canonical intersection with the pulled-back $(d-1)$-st power of R. Let L_0 be the pullback of L from either slot. It is nef, Cartier, and of full nef dimension. We claim

$$K_{V^*} + L_0 \text{ big.} \tag{25}$$

The pushforward $N_0 = (p_0)_* L_0$ is pseudo-effective on A_0, and is therefore nef. Indeed translations make the pseudo-effective and nef cones of an abelian variety coincide. If N_0 were not big, the standard description of nef line bundles on abelian varieties would give a positive-dimensional abelian subvariety on whose translates N_0 is numerically trivial. One can see this from the semipositive Hermitian form of the numerical class: its kernel is rational for the lattice, since the alternating form of the integral Cartier class is integral, and therefore defines an abelian subvariety. A general translate meets the image of the exceptional locus of p_0 in codimension at least two, if at all. Complete-intersection curves in that translate can thus be chosen through very general points and avoiding this image. Their lifts have L_0-degree zero, contradicting full nef dimension. Hence N_0 is big.

By the negativity lemma,

$$p_0^* N_0 - L_0 = E_0 \geq 0$$

is exceptional. The canonical divisor of V^* has an effective representative E_K with positive coefficient along every p_0-exceptional divisor, because the target is smooth with trivial canonical bundle. Choose a rational $0 < u < 1$ sufficiently small that $E_K - uE_0 \geq 0$. Then

$$K_{V^*} + uL_0 \sim_{\mathbb{Q}} up_0^* N_0 + (E_K - uE_0)$$

is big. Adding the nef divisor $(1-u)L_0$ proves (25).

We may now apply Lemma 2.1 in dimension d on V^*, with testing class R and empty boundary. The bounds (23) give both the top intersection bound and the canonical ratio, while $R - rL_0$ is nef and (25) supplies bigness. This final contradiction excludes the correspondence case.

Conclusion of the proof of Proposition 3.1. The alternatives for every moving base component have been exhausted: first its slice fibres, then components of dimension less than d, sweeping branch divisors, and finally the fixed covers. In each case the contradiction comes from Lemma 2.1 with constants uniform in t. The finite cover lists were used only

for one fixed t, and their sizes do not enter the degree bounds. Thus failures of (13) cannot occur along a sequence $t \to 0$. This proves the proposition, including the additional open conditions on the chosen point. □

4. Proof of Lemma 1.1

Proof. Fix $s = q$ supplied by Proposition 3.1. For every $n \gg 0$, choose $z \in U$ such that

$$\varepsilon(R_{n,s}; z) > 2d + 2.$$

Choose a rational number a strictly between these two numbers. On the blowup $b : \widetilde{Z} \to Z$ at z, with exceptional divisor G, the divisor $b^*R_{n,s} - aG$ is ample. Serre vanishing and the restriction sequence imply that, for all sufficiently large divisible k, $|kR_{n,s}|$ separates $(ka-1)$-jets at z. For such k, $ka - 1 \geq (2d+2)k$.

For one such k_0, this jet surjectivity is an open rank condition. It therefore holds on a nonempty open subset of U. Products of the degree-k_0 sections separate $\ell(2d+2)k_0$-jets in degree ℓk_0 for every $\ell \geq 1$. Indeed, every monomial of total degree at most $\ell(2d+2)k_0$ is a product of ℓ monomials of total degree at most $(2d+2)k_0$, and the lifted products span the jet quotient. This proves the assertion at a very general point and for all sufficiently divisible k. □

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